

$$\mathcal{G}_h(\mathbf{z}) = \mathbf{z} + h\mathbf{f}(\mathbf{z}),$$

and we have

$$\mathbf{z}_{n+1} = \mathcal{G}_h(\mathbf{z}_n), \quad n = 0, 1, 2, \dots \quad (2.2)$$

with $\mathbf{z}_0 = \boldsymbol{\xi}$. The points $\{\mathbf{z}_n\}$ are intended to be approximations to the values of the solution. The obvious question is this: how good an approximation is $\mathbf{z}_n$ of $\mathbf{z}(nh)$?

Let the approximate solution vectors at successive timesteps be $\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_v$ where $vh = \tau$. We assume that τ , the length of the time interval, is fixed, and v is an integer parameter representing the total number of timesteps. In order to improve the quality of the approximation, the parameter v may be increased, as the stepsize is proportionately decreased. The error at step n is defined by $e_n = \|\mathbf{z}_n - \mathbf{z}(t_n)\|$, where $t_n = nh$; it clearly depends on h . With these definitions one can prove the following result, which is typical of the sorts of results that are available for numerical methods for initial value problems:

Theorem 2.1 *Let $\mathcal{D}$ be a bounded, open region in $\mathbb{R}^m$ such that $\mathbf{f} : \mathcal{D} \rightarrow \mathbb{R}^m$ is continuously differentiable. Let $\boldsymbol{\xi}$ be an interior point of $\mathcal{D}$ and suppose the initial value problem (2.1) has a unique solution that remains in $\mathcal{D}$ for $t \in [0, \tau]$. Then there exists a constant $C(\tau) > 0$ such that for sufficiently large $v \in \mathbb{N}$ the numerical solution $\mathbf{z}_n$ remains in $\mathcal{D}$ for $n = 0, 1, \dots, v$, where $hv = \tau$, and, moreover, the maximum global error in Euler's method satisfies*

$$\bar{e} := \max_{0 \leq n \leq v} e_n \leq C(\tau)h.$$

In short, the error in the approximation obtained on $[0, \tau]$ is reduced in direct proportion to the number of steps taken to cover this interval. Another way to say this is that $\bar{e} \propto h$, or, using the order notation, $\bar{e} = \mathcal{O}(h)$. Because the global error is of order h^r , where $r = 1$, we say that Euler's method is a *first order* method, or that it *converges with order $r = 1$* .

2.1.1 Example: The Trimer

Let us test Euler's method on the trimer model (formulated as a system in $\mathbb{R}^4$). Fixing initial values ($\mathbf{q}_0 = (0.5, 1.0); \mathbf{p}_0 = (0.1, 0.1)$) we solve the trimer on $[0, 4]$ using the Euler method first with 40 time-steps of length 0.1, next using 80 time-steps of length 0.05 and then with 160 steps of size 0.025, etc. Each calculation results in a different “discrete trajectory.” The solutions are graphed in the xy projection in the left panel of Fig. 2.2 with line segments connecting the points; these piecewise linear paths take on the appearance of smooth curves as the stepsize is reduced.